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Specific heat of the R_3Co (R = heavy rare earth or Y) compounds

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Specific heat measurements were performed in wide temperature range T = 2–300 K for the R_3Co (R = Er–Gd, Y) compounds. Using values of the Sommerfeld coefficient and Debye temperature estimated for Y_3Co compound, the Debye temperatures, phonon and conduction electron contributions as well as magnetic contribution to the specific heat were estimated for all other R_3Co compounds.

1. Introduction The neutron diffraction study performed for some of the R_3M compounds, where R is rare earth and M=Ni or Co, has shown that only rare earth atoms carry magnetic moment in those compounds while transition metal remains non-magnetic [1, 2]. The exchange interactions between rare earth ions are realised by means of the hybridised 3d–4d electrons when R=Y or 3d–5d electrons when R is a rare earth element [3]. The low symmetry orthorhombic crystalline structure in which R atoms occupy two nonequivalent sites leads to crystal field effects increasing. The coexistence of the strong exchange interactions and large magnetic anisotropy results in coexistence of the ferro- and antiferromagnetic properties in the compounds. Comparatively low temperatures of the magnetic ordering and high intensity of the saturation magnetization make the compounds prospective for new magnetic refrigerators designing. Hence, the methodical investigations of these compounds are interesting for substantial physical processes understanding as well as for application problem solving.

However, the specific heat study of the compounds has sporadic character. Until now, the specific heat investigations were carried out only for a few R_3M compounds, including the Er_3Ni [4], Gd_3Co [5] and Dy_3Co [6], in rather narrow ranges of temperature. The purpose of the work was to investigate and analyze the specific heat for the number of the R_3Co (R is heavy rare earth or Y) compounds performed in wide range of temperatures.

2. Experimental All samples used in the work were obtained by high-frequency melting of the constituent metals in a purified argon atmosphere under pressure of 50 kPa. The X-ray diffraction measurements on the powdered samples confirmed that they were single-phase. The specific heat measurements were performed using Quantum Design PPMS Heat Capacity System in temperature range of 2–300 K.

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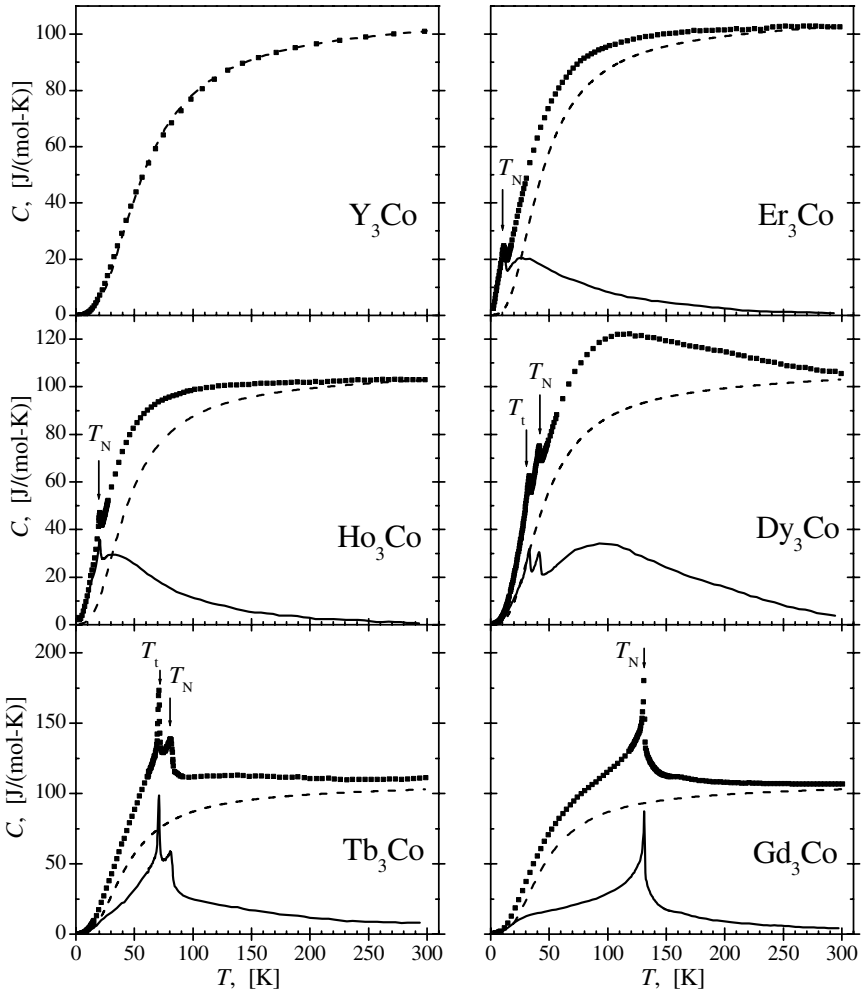


Fig. 1 The specific heat of the R_3Co compounds (squares). The dashed lines show conductivity electrons and phonon contributions. The solid lines show the magnetic contributions to the total specific heat. The arrows show the magnetic transition temperatures.

3. Results and discussion The temperature dependencies of the specific heat $C_p(T)$ for R_3Co (R=Y, Er, Ho, Dy, Tb, and Gd) compounds are given in Fig. 1.

The Y_3Co compound exhibits Pauli-type paramagnetism. The sharp λ-like maximums on the specific heat curves corresponding to the magnetic ordering transition T_N are observed for the other R_3Co compounds. The Néel temperatures obtained from the specific heat measurements (see Table 1) are in good agreement with previous data [7]. The peaks at $T_t = 32.1$ and 70.5 K observed for the Dy_3Co and Tb_3Co compounds, respectively, can be associated with phase transitions from one antiferromagnetic state to another. All mentioned above anomalies are typical for second order phase transitions.

Table 1 The Néel and Debye temperatures for the R_3Co compounds.

	Y_3Co	Er_3Co	Ho_3Co	Dy_3Co	Tb_3Co	Gd_3Co
T_N, [K]	–	11.2	20.4	40.8	80.4	129.0
Θ_D, [K]	234	176	177	179	180	181

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The measurement of the specific heat of the isostructural nonmagnetic Y_3Co compound was performed to the isolate conduction electrons and phonon contributions from the total specific heat. The fitting was performed according to formula:

$$C_{\mathrm{el+ph}} = \gamma T + 9NR(T/\Theta_{\mathrm{D}})^3 \int_0^{\Theta_{\mathrm{D}}/T} \frac{x^4 e^x}{(e^x - 1)^2}\,\mathrm{d}x\,, \tag{1}$$

where the first term represents electronic specific heat and the second one corresponds to the phonon contribution of the Debye model, $N = 4$ is the number of atoms per formula unit and R is a molar gas constant. The specific heat of the Y_3Co follows Eq. (1) satisfactory with a Debye temperature $\Theta_{\mathrm{D}} = 234$ K and Sommerfeld coefficient $\gamma \sim 15$ mJ/(mol K^2) for the whole range of temperatures 2–300 K.

The Debye temperatures for other R_3Co compounds were calculated by methods offered in [8], taking into account the different atom masses and effective Debye temperatures of the constituent elements [9]. The Debye temperatures obtained by various methods are with comparable 5% accuracy. The values which we have used for further analysis are collected in Table 1. The $C_{\mathrm{el-ph}}(T)$ dependencies, calculated according to Eq. (1) for each of the investigated compounds, are shown in Fig. 1 by dashed lines.

The magnetic contribution to the total specific heat $C_{\mathrm{m}}(T)$ was estimated by subtracting the calculated electron-phonon contribution $C_{\mathrm{el-ph}}(T)$ from the measured specific heat $C_{\mathrm{P}}(T)$ for each of the investigated compounds. The wide smooth maximums observed on $C_{\mathrm{m}}(T)$ for the Er_3Co, Ho_3Co and Dy_3Co compounds at temperatures above T_{N} as well as small anomaly of the magnetic specific heat for Gd_3Co at around 30 K can be associated with Schottky like anomaly caused by splitting of the R^{3+} ions ground state multiplet due to crystal field effect. The Er, Ho and Dy ions have large orbital moments ($L = 6$, 6 and 5, respectively) and, therefore, not only low symmetry orthorhombic structure but also the anisotropy of the rare earth element give rise to the crystal field effect. The Gd ion is isotropic ($L = 0$) and for the Gd_3Co compound the Schottky like effect is significantly lower. We suppose, that the Schottky like anomaly was not observed for the Tb_3Co compound because strong anomalies corresponding to the magnetic ordering transitions are close to the temperatures where the Schottky like anomaly could be expected and it is possible, that the Schottky like effect is covered by the magnetic ordering anomalies.

The magnetic part of the entropy $S_{\mathrm{m}}(T)$ was calculated by integrating $C_{\mathrm{m}}(T)/T$ dependence for each of the investigated compounds and the results are shown in Fig. 2. The facts that the entropy does not reach theoretical maximum value $S_{\max} = 3R \ln(2J + 1)$, where J is total angular momentum of a rare earth ion, at Néel temperatures in all investigated compounds as well as that noticeable magnetic contribution to

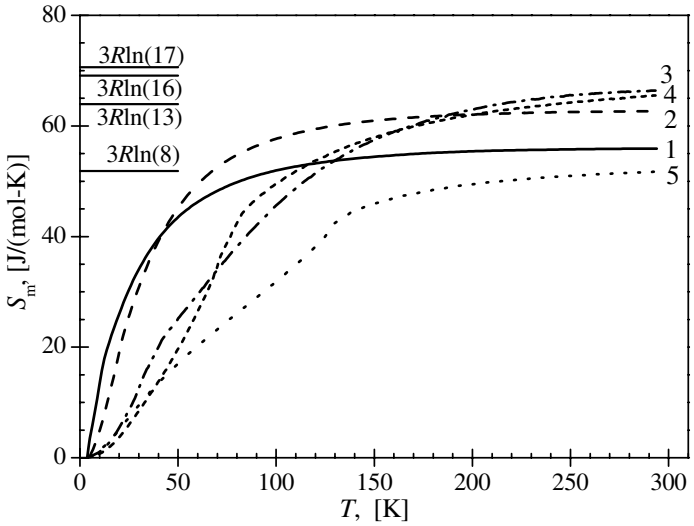


Fig. 2 The magnetic entropy for (1) Er_3Co $\{J_{\mathrm{Er}} = 15/2\}$, (2) Ho_3Co $\{J_{\mathrm{Ho}} = 8\}$, (3) Dy_3Co $\{J_{\mathrm{Dy}} = 15/2\}$, (4) Tb_3Co $\{J_{\mathrm{Tb}} = 6\}$ and (5) Gd_3Co $\{J_{\mathrm{Gd}} = 7/2\}$ compounds. Short lines on the left side show the theoretical values $3R(2J + 1)$ of the entropy.

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specific heat persists up to at least $2T_N$ can be an evidence that spin fluctuations in d-electron subsystem plays significant role far above magnetic ordering temperature for these compounds.

For the Gd_3Co compound the magnetic entropy tends to saturation at $T > 270$ K and reaches S_{max}. For the Er_3Co and Ho_3Co compounds the entropy tends to saturation at $T > 200$ K but does not reach S_{max} what can be a result of strong change in energy level system of the Er^{3+} and Ho^{3+} ions ground state multiplet due to Schottky like effect. However, extensive analysis, considering local anisotropy of the rare earth ions and the effect of the crystal field on the total angular momentum of the rare earth ions, should be performed in order to explain the anomalies of the entropy behavior.

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